

Genome-wide Association Studies

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Roadmap of STAT/BIOF/GSAT540 2026

data generation followed by *inference* or *validation*

- ③ Formulating biological questions
- ④ Hypothesis testing
- ⑤ RNA-seq data
- ⑥ Inference focusing on RNA-seq
- ⑦ Experimental design
- ⑧ Single-cell genomics
- ⑨ Matrix factorization
- ⑩ **Genome-wide association studies** (Today)
- ⑪ Multiple hypothesis testing in Genomics

Learning objectives

- Background knowledge of statistical genetics
- Biological and statistical intuitions

Today's lecture

- 1 Mapping disease-specific locations in Genome
- 2 A toy example GWAS
- 3 How GWAS can be interpreted wrongfully
- 4 Combining evidence from multiple studies

Genetics Recap: What is a gene?

- Gene = a unit of heredity copied from a parent to an offspring
- Key discoveries:
 - Gregor Mendel's pea experiments,
 - Morgan's fruit fly work.

A gene can take many forms...

Allele

- A different form of a gene
- A Greek word “allos,” $\alpha\lambda\lambda\omicron\varsigma$, meaning “other”

Variant and locus

- A specific region of the genome differs across two or more genomes
- A result of mutation
- A locus: a location where many variants lie (plural: loci).

“We found *A* allele for this genetic variant as an effect allele.”

“We found a hundred variants within the locus of *APOE*.”

“We identified ten loci associated with cancer.”

Ploidy and bi-allelic variants

Ploidy

- The number of copies of chromosomes within a cell/organism
- Haploid: one copy
- Diploid: two copies

Biallelic variant

- bi + allelic
- Two forms for a variant
 - **Reference** (more frequent)
 - **Alternative** (less frequent)

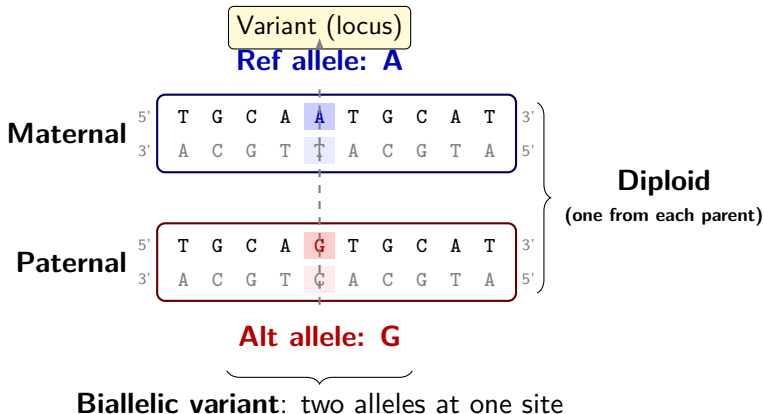
Minor allele frequency

- For a biallelic variant, the **minor allele** is the less frequent (alternative) allele in a population
- **Minor Allele Frequency (MAF)**: the proportion of haplotypes carrying the minor allele

$$\text{MAF} = \frac{\# \text{ minor alleles}}{\# \text{ total alleles (haplotypes)}}$$

- In a diploid genome with n individuals: total alleles = $2n$
- Common variant: $\text{MAF} \geq 1\%$; rare variant: $\text{MAF} < 1\%$

Let's build up what we know...



Single nucleotide polymorphism

Polymorphism

- Poly + morph
- Occurrence of different forms

SNP

- Single Nucleotide Polymorphism
- A place in the genome where people differ by a single base pair

We pronounce SNP “snip” in North America.

Homozygotes vs. Heterozygotes in diploid genome

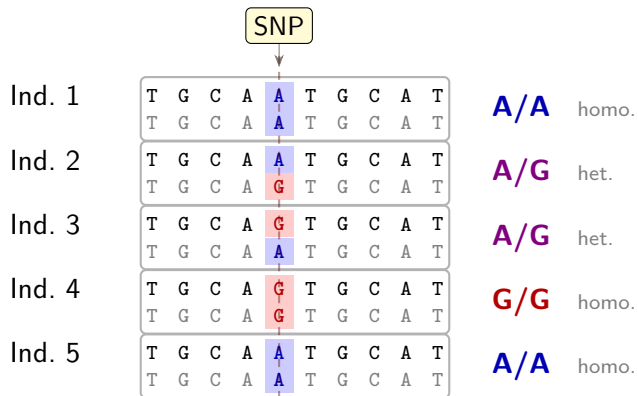
Zygote

- A fertilized egg cell
- Formed by the fusion of two gametes (sperm + egg)
- Contains two copies of each chromosome (diploid)

Homo- vs. heterozygote

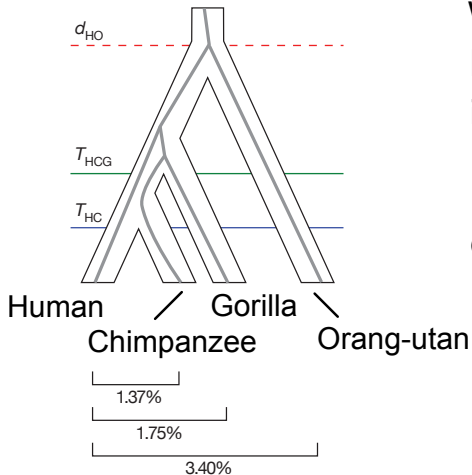
- Homozygote: same allele on both copies (e.g., A/A or G/G)
- Heterozygote: different alleles (e.g., A/G)

Single nucleotide polymorphism (just 5' → 3')



Genetics is described in its own terminology.

How and why can we study human genetics?



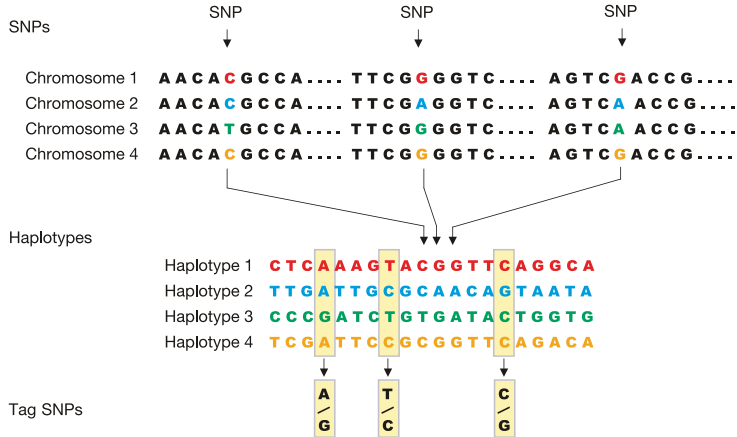
We can study human genetics because humans share a 99% identical genome.

GWAS is only possible because:

- Although human DNA sequence $n \approx 3.2 \times 10^9$ bp,
- luckily # SNPs $n \approx 4-5 \times 10^6$

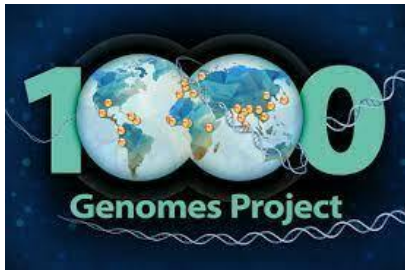
Scally et al. (2012)

International HapMap to identify “tag” SNPs



International HapMap Consortium (2003)

Human genetics data — The 1000 Genomes Project



Let's download the 1KG data using bigsnpr library.

```
library(bigsnp)  
download_1000G("../data/genotype/")
```

<https://www.internationalgenome.org/>

- International consortium
- Goal: find common genetic variants with frequencies of at least 1%
- The project planned to sequence each sample to 4x genomic coverage.

A big matrix data set with row and column data

Normally, we have a set of three files (PLINK format):

- ❶ `../data/genotype/1000G_phase3_common_norel.bed`
- ❷ `../data/genotype/1000G_phase3_common_norel.bim`
- ❸ `../data/genotype/1000G_phase3_common_norel.fam`

What are the columns? genetic variants/SNPs

```
fread(.bim.file, nrow=8)
```

##	V1	V2	V3	V4	V5	V6
##	<int>	<char>	<int>	<int>	<char>	<char>
## 1:	1	rs2185539	0	566875	T	C
## 2:	1	rs3131972	0	752721	G	A
## 3:	1	rs12184325	0	754105	T	C
## 4:	1	rs3131969	0	754182	G	A
## 5:	1	rs3131967	0	754334	C	T
## 6:	1	rs3131962	0	756604	G	A
## 7:	1	rs1048488	0	760912	T	C
## 8:	1	rs115991721	0	767096	G	A

- 1 Chromosome code
- 2 Variant identifier
- 3 Position in morgans or centimorgans (dummy '0')
- 4 Base-pair coordinate (1-based; limited to $2^{31}-2$)
- 5 Allele 1 (corresponding to clear bits in .bed; usually minor)
- 6 Allele 2 (corresponding to set bits in .bed; usually major)

<https://www.cog-genomics.org/plink/1.9/formats>

What are the rows? samples/individuals/subjects

```
fread(.fam.file, nrows=7)
```

##	V1	V2	V3	V4	V5	V6
##	<int>	<char>	<int>	<int>	<int>	<int>
## 1:	0	HG00096	0	0	1	-9
## 2:	0	HG00097	0	0	2	-9
## 3:	0	HG00099	0	0	2	-9
## 4:	0	HG00100	0	0	2	-9
## 5:	0	HG00101	0	0	1	-9
## 6:	0	HG00102	0	0	2	-9
## 7:	0	HG00103	0	0	1	-9

- 1 Family ID ('FID')
- 2 Within-family ID ('IID'; cannot be '0')
- 3 Within-family ID of father ('0' if father isn't in dataset)
- 4 Within-family ID of mother ('0' if mother isn't in dataset)
- 5 Sex code (1: male, 2: female, 0: unknown)
- 6 Phenotype value¹ (1: control, 2: case, '-9': missing)

<https://www.cog-genomics.org/plink/1.9/formats>

¹not much used

A variant $\approx$ SNP (throughout the course)

example

1 rs12184325 0 754105 T C

- 1 Chromosome
- 2 Variant
- 3 Morgan (ignore)
- 4 Base-pair coordinate
- 5 Allele 1
- 6 Allele 2

A variant $\approx$ SNP (throughout the course)

example

1 rs12184325 0 754105 T C

① Chromosome: **1**

A variant $\approx$ SNP (throughout the course)

example

1 rs12184325 0 754105 T C

- 1 Chromosome: **1**
- 2 Variant identifier: **rs12184325**, a unique ID used in dbSNP; can be anything unique; it can change depending on the dbSNP human genome build (don't rely on it).

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1 rs12184325 0 754105 T C

- 1 Chromosome: **1**
- 2 Variant identifier: **rs12184325**, a unique ID used in dbSNP; can be anything unique; it can change depending on the dbSNP human genome build (don't rely on it).
- 3 A unit for measuring genetic linkage (just put the dummy value "0")
- 4 Genomic position (bp) within each chromosome

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- 1 Chromosome: **1**
- 2 Variant identifier: **rs12184325**, a unique ID used in dbSNP; can be anything unique; it can change depending on the dbSNP human genome build (don't rely on it).
- 3 A unit for measuring genetic linkage (just put the dummy value "0")
- 4 Genomic position (bp) within each chromosome
- 5 **A1**: If this allele (haplotype) is *T*, we use **0**
- 6 **A2**: If this allele (haplotype) is *C*, we use **1**

A variant $\approx$ SNP (throughout the course)

example

```
1 rs12184325 0 754105 T C
```

- How many haplotypes? A conventional way to code "genotype" is to sum them up (additive effect); there are several other options (dominance, heterozygous).

A variant $\approx$ SNP (throughout the course)

example

```
1 rs12184325 0 754105 T C
```

- How many haplotypes? A conventional way to code "genotype" is to sum them up (additive effect); there are several other options (dominance, heterozygous).
- How do you code "TT"?
- How do you code "CC"?
- How do you code "CT"?

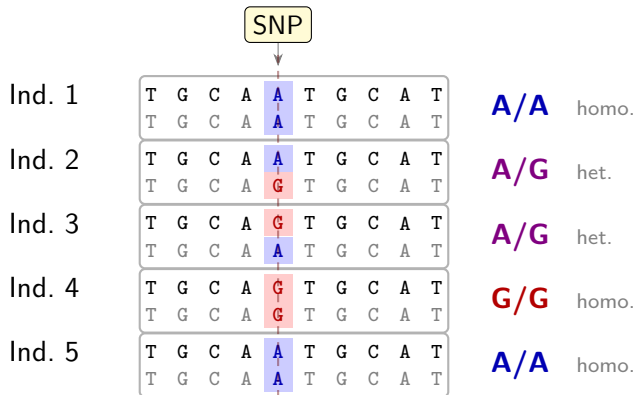
A variant $\approx$ SNP (throughout the course)

example

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1 rs12184325 0 754105 T C
```

- How many haplotypes? A conventional way to code "genotype" is to sum them up (additive effect); there are several other options (dominance, heterozygous).
- How do you code "TT"? **0**
- How do you code "CC"? **2**
- How do you code "CT"? **1**

Recap: Single nucleotide polymorphism (just 5' → 3')



Pop quiz: Statistical Genetics 101

Question 1

In GWAS, genotypes are typically coded as 0, 1, or 2. What does a genotype value of “1” represent under the additive model?

- Ⓐ Homozygous for the reference allele
- Ⓑ Heterozygous
- Ⓒ Homozygous for the alternative allele
- Ⓓ Missing data

Pop quiz: Statistical Genetics 101

Question 2 In a standard genotype matrix with n individuals and p SNPs, what are the dimensions?

- ☐ A) n rows (individuals) $\times$ p columns (SNPs)
- ☐ B) p rows (SNPs) $\times$ n columns (individuals)
- ☐ C) n rows (SNPs) $\times$ p columns (individuals)
- ☐ D) The matrix must be square ($n = p$)

Pop quiz: Statistical Genetics 101

Question 3 What is the Minor Allele Frequency (MAF)?

- Ⓐ The frequency of the most common allele in the population
- Ⓑ The frequency of the least common allele in the population
- Ⓒ The frequency of heterozygous individuals
- Ⓓ The frequency of missing genotypes

How these genotypes are instantiated in 1KG data

By calling `snp_readBed(.bed.file)`, we can convert the “BED”-formatted data to an “RDS” file for faster access. Later, we need to “attach” that RDS.

```
data <- snp_attach(.bk.file)
str(data, max.level=1, strict.width = "cut")

## List of 3
## $ genotypes:Reference class 'FBM.code256' [package "bigstatsr"] with 16 fields
## ..and 26 methods, of which 12 are possibly relevant
## $ fam      :'data.frame': 2490 obs. of 6 variables:
## $ map      :'data.frame': 1664852 obs. of 6 variables:
## - attr(*, "class")= chr "bigSNP"
```

How these genotypes are instantiated in 1KG data

```
dim(data$genotypes)
```

```
## [1] 2490 1664852
```

```
dim(data$fam)
```

```
## [1] 2490 6
```

```
dim(data$map)
```

```
## [1] 1664852 6
```

- How many rows and columns?
- What are the rows and columns?

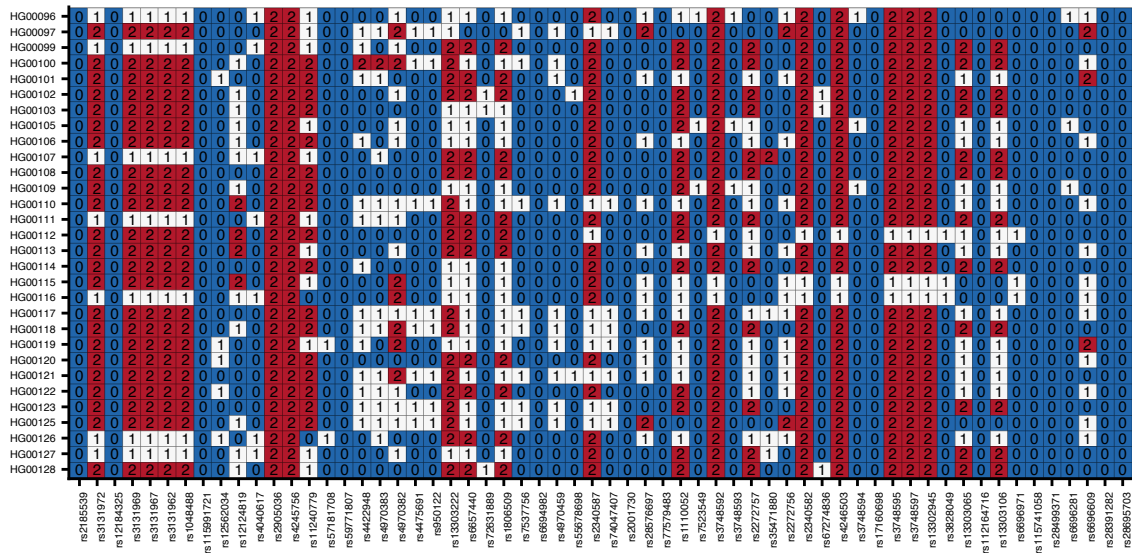
How these genotypes are instantiated in 1KG data

A genotype/dosage matrix:

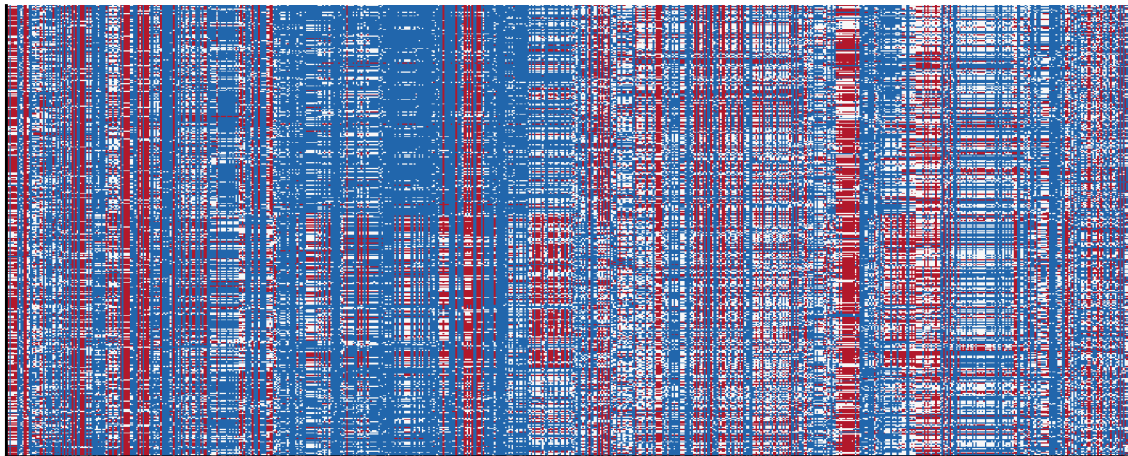
```
data$genotypes[1:5, 1:10]
```

##		[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]
##	[1,]	0	1	0	1	1	1	1	0	0	0
##	[2,]	0	2	0	2	2	2	2	0	0	0
##	[3,]	0	1	0	1	1	1	1	0	0	0
##	[4,]	0	2	0	2	2	2	2	0	0	1
##	[5,]	0	2	0	2	2	2	2	0	1	0

A tiny fraction of the full genotype matrix



Just a bit more: Do you see any patterns?



We will get back to the structured patterns.

Goal: mapping disease-specific locations in Genome

GWAS

- Input:

- A genotype matrix X ($n \times p$), where $X_{ij} \in \{0, 1, 2\}$
- A phenotype vector $\mathbf{y}$ ($n \times 1$)

- Optimization:

$$\mathbf{y} = \mathbf{x}_j \beta_j + \epsilon, \text{ for each } j$$

- Output:

- Estimate an effect size (coefficient) β_j ($j \in 1, \dots, p$)

- Testing $H_0 : \beta_j = 0$ vs. $H_1 : \beta_j \neq 0$.

Goal: mapping disease-specific locations in Genome

GWAS

- Input:
 - A genotype matrix X ($n \times p$), where $X_{ij} \in \{0, 1, 2\}$
 - A phenotype vector y ($n \times 1$); it can be anything...
- Output:
 - Estimate a coefficient (effect size) β_j ($j \in 1, \dots, p$)
- Testing $H_0 : \beta_j = 0$ vs. $H_1 : \beta_j \neq 0$.
- Report p-values
- Is it still straightforward after seeing the X matrix—so direly structured?

What are the differences between DEG and GWAS?

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So many variants, much fewer samples/individuals

What are the differences between DEG and GWAS?

So many variants, much fewer samples/individuals

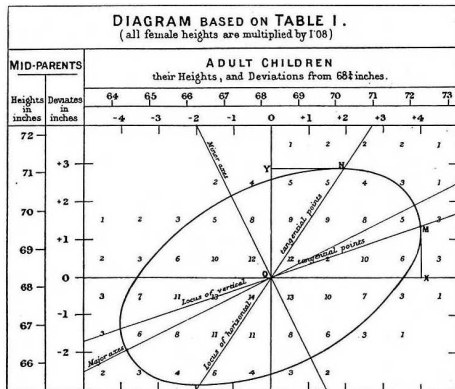
Expression $\sim$ phenotype vs. phenotype $\sim$ genetic variants

What are the differences between DEG and GWAS?

So many variants, much fewer samples/individuals

Goal: mapping a genetic variant $\rightarrow$ a phenotype

One more concept: heritability



Sir Francis Galton's diagram

https://en.wikipedia.org/wiki/Regression_toward_the_mean

- How do we know a certain trait is heritable?
- What fraction of the variance in the offspring can be explained by the parents?

A missing heritability problem

NEWS FEATURE PERSONAL GENOMES

NATURE | Vol 456 | 6 November 2008



- We knew some traits are heritable...
- However, we could only explain very little by GWAS...

Manolio et al. (2009)

How much of trait variation is genetic?

- Not all traits are equally heritable
- Height: ~80% heritable
- BMI: ~40-70% heritable
- Complex diseases: varies widely

How much of trait variation is genetic?

- Not all traits are equally heritable
- Height: ~80% heritable
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GWAS perspective:

- Low heritability → weak association signals
- High heritability but few GWAS hits → “missing heritability” problem
- Heritability estimates guide study design and power calculations

Broad-sense heritability for a certain trait

$$V_P = V_G + V_E$$

- V_P : Total phenotypic variance (what we observe)
- V_G : **Genetic variance** (all genetic effects)
- V_E : Environmental variance (everything else)

Broad-sense heritability H^2

$$H^2 = \frac{V_G}{V_P} = \frac{V_G}{V_G + V_E}$$

Proportion of phenotypic variance due to **all** genetic factors.

Narrow-sense heritability

$$V_G = \underbrace{V_A}_{\text{our interest}} + V_D + V_I$$

- V_A : **Additive** genetic variance (linearity)
- V_D : **Dominance** variance (non-linearity)
- V_I : **Epistatic** variance (interactions)

Narrow-sense heritability h^2

$$h^2 = \frac{V_A}{V_P} = \frac{V_A}{V_G + V_E}$$

Proportion of variance due to **additive**/linear genetic effects only.

Most GWAS assume an additive model

For each variant j :

$$y_i = \beta_0 + \beta_j x_{ij} + \epsilon_i$$

where $x_{ij} \in \{0, 1, 2\}$ counts alleles (genotypes).

Most GWAS assume an additive model

For each variant j :

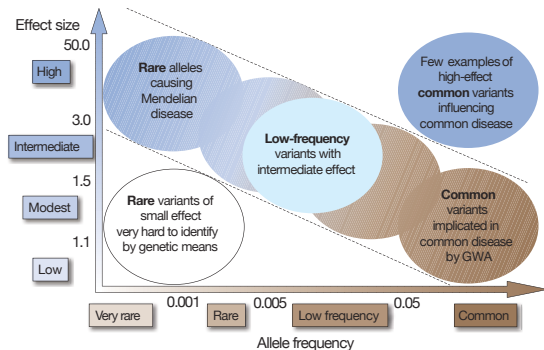
$$y_i = \beta_0 + \beta_j x_{ij} + \epsilon_i$$

where $x_{ij} \in \{0, 1, 2\}$ counts alleles (genotypes).

Implications:

- GWAS primarily captures the V_A (additive variance) component
- Narrow-sense heritability h^2 is the relevant measure for GWAS
- h^2 sets an upper bound on GWAS explanatory power

Practical implications with respect to MAF



Manolio et al. (2009)

- Low heritability h^2 per SNP generally implicates
 - high MAF (common)
 - low effect size
 - E.g., each SNP explains a very tiny fraction height.
- High heritability h^2 per SNP generally implicates
 - low MAF (rare)
 - high effect size
 - E.g., very small number of rare variants can explain cancer.

Heritability in our simulation

A typical simulation in R:

```
simulate.pgs <- function(X, W, h2, pve.w, n.causal, n.traits = 1) {  
  # ... genetic effects ...  
  y.obs <- y.true * sqrt(h2) + y.pop * sqrt(pve.w)  
  y.obs <- y.obs + y.err * sqrt(1 - h2 - pve.w)  
  # ...  
}  
  
.sim <- simulate.pgs(X, W, h2=.3, pve.w=0, n.causal=3)
```

- $h^2 = 0.3$: SNP heritability is 30% (3 causal SNPs contribute to genetic variance)
- $pve.w$: Proportion of variance from population structure (confounding)
- $1 - h^2 - pve.w$: Residual environmental variance

This simulates a trait with moderate SNP heritability (30%).

A wrap-up quiz

Why does GWAS primarily estimate narrow-sense heritability (h^2) rather than broad-sense heritability (H^2)?

- Ⓐ GWAS uses additive genetic models for each SNP
- Ⓑ Dominance effects are always negligible
- Ⓒ Epistatic effects are easy to detect
- Ⓓ H^2 is not a valid concept for complex traits

A wrap-up quiz

What is the relationship between broad-sense heritability (H^2), narrow-sense heritability (h^2), and SNP heritability (h^2_{SNP})?

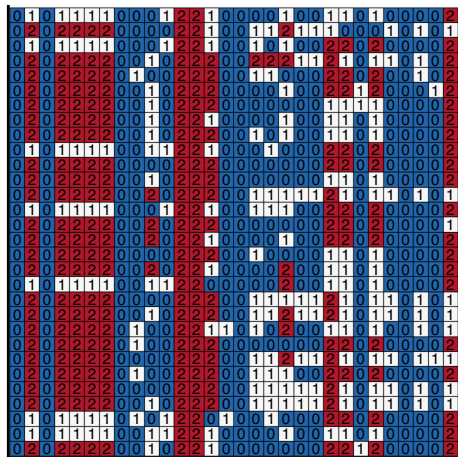
- A) $H^2 \leq h^2 \leq h^2_{\text{SNP}}$
- B) $h^2_{\text{SNP}} \leq h^2 \leq H^2$
- C) $h^2 \leq H^2 \leq h^2_{\text{SNP}}$
- D) They are all equal

Today's lecture

- 1 Mapping disease-specific locations in Genome
- 2 A toy example GWAS**
- 3 How GWAS can be interpreted wrongfully
- 4 Combining evidence from multiple studies

GWAS for “Obsessive ggplot Disorder”

A genotype matrix X ($X_{ij} \in \{0, 1, 2\}$):



A vector of phenotypes Y ($Y_i = 1$ if case vs. $Y_i = 0$ if control)

```
table(Y)
```

```
## Y
```

```
##      0      1
```

```
## 1272 1218
```

Goal: Find locations (columns of X) where a genotype x_j is associated with a phenotype y

²an illustration purpose

Let's do GWAS to find the disease-associated variants

A variant-by-variant association t-test for a variant j :

$$H_0 : \mathbb{E}[X_{ij}|Y_i = 1] = \mathbb{E}[X_{ij}|Y_i = 0] \quad \text{vs.} \quad \mathbb{E}[X_{ij}|Y_i = 1] \neq \mathbb{E}[X_{ij}|Y_i = 0]$$

- The average genotype is the same between the case and control under the null.

```
j <- 1  
t.test(X[Y == 0, j], X[Y == 1, j])
```

```
##  
## Welch Two Sample t-test  
##  
## data: X[Y == 0, j] and X[Y == 1, j]  
## t = 0.16273, df = 2468.6, p-value = 0.8707  
## alternative hypothesis: true difference in means is not equal to 0  
## 95 percent confidence interval:  
## -0.02015587 0.02380397  
## sample estimates:  
## mean of x mean of y  
## 0.06996855 0.06814450
```

Can we calculate GWAS statistics more efficiently?

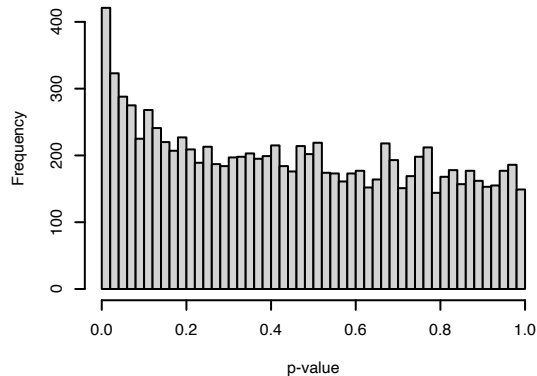
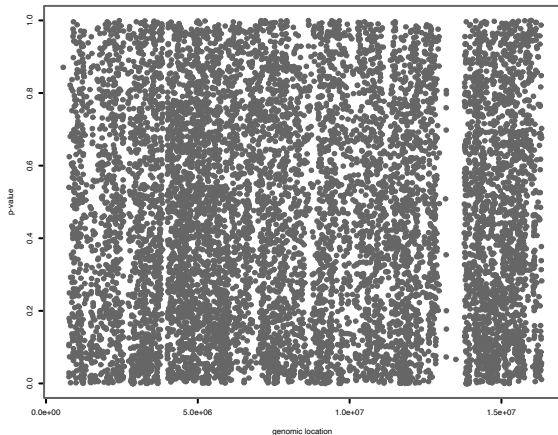
```
## library(matrixTests)
.gwas <- col_t_welch(X[Y == 0, ], X[Y == 1, ])
```

- For each pair of columns in the two matrices, the function performs t-test and summarize all the results into a list of vectors.

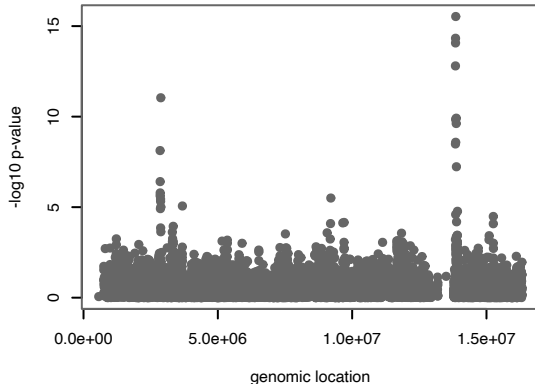
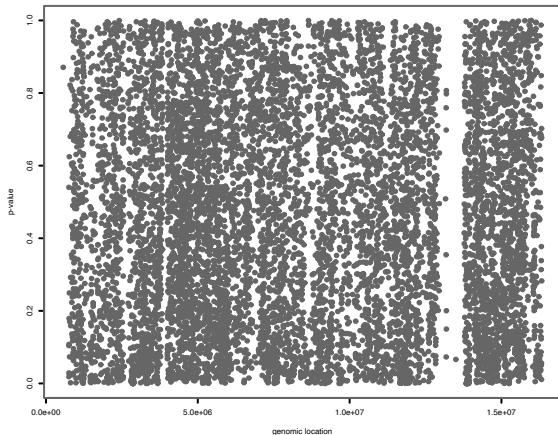
```
names(.gwas)
```

```
## [1] "obs.x"      "obs.y"      "obs.tot"    "mean.x"     "mean.y"
## [6] "mean.diff"  "var.x"      "var.y"      "stderr"     "df"
## [11] "statistic"  "pvalue"     "conf.low"   "conf.high"  "mean.null"
## [16] "alternative" "conf.level"
```

Manhattan plot: A quick summary of all the GWAS p-values

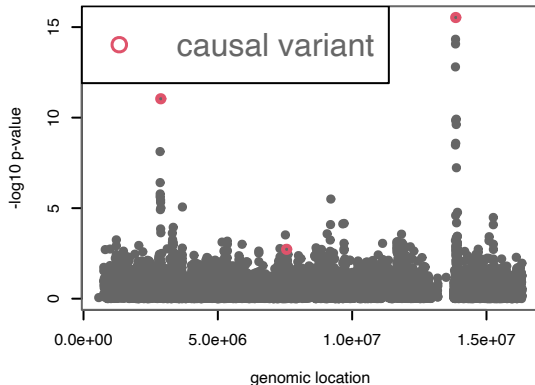
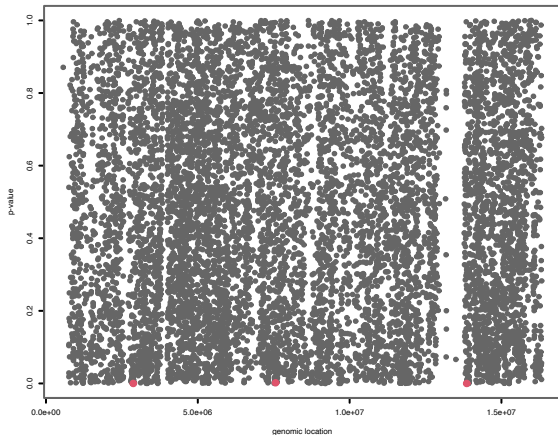


Manhattan plot: A quick summary of all the GWAS p-values



Which one is more informative?

Manhattan plot: A quick summary of all the GWAS p-values



Which one is more informative?

Where are the variants significantly

Multiple hypothesis adjustment (next lecture)

We have many p-values, namely, $p_1, p_2, \dots, p_{10000}$.

- ① In GWAS, we usually take a very conservative stance. So, we adjust the nominal p-values by Bonferroni adjustment.

```
padj <- p.adjust(.gwas$pvalue)  
sum(padj < .05)
```

```
## [1] 18
```

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We have many p-values, namely, $p_1, p_2, \dots, p_{10000}$.

- ① In GWAS, we usually take a very conservative stance. So, we adjust the nominal p-values by Bonferroni adjustment.
- ② What will be the adjusted p-value for p_j ?

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- ③ $p_j^{\text{adj}} \leftarrow \min\{1, p_j \times 10000\}$

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Multiple hypothesis adjustment (next lecture)

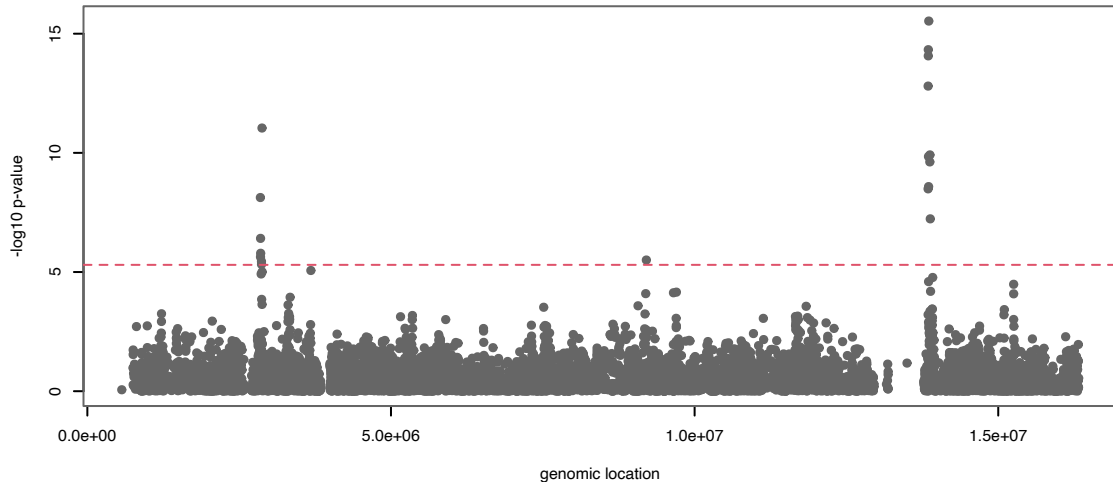
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- ① In GWAS, we usually take a very conservative stance. So, we adjust the nominal p-values by Bonferroni adjustment.
- ② What will be the adjusted p-value for p_j ?
- ③ $p_j^{\text{adj}} \leftarrow \min\{1, p_j \times 10000\}$
- ④ $p_j < \alpha/10000$ where 10000 is the total number of hypothesis tests

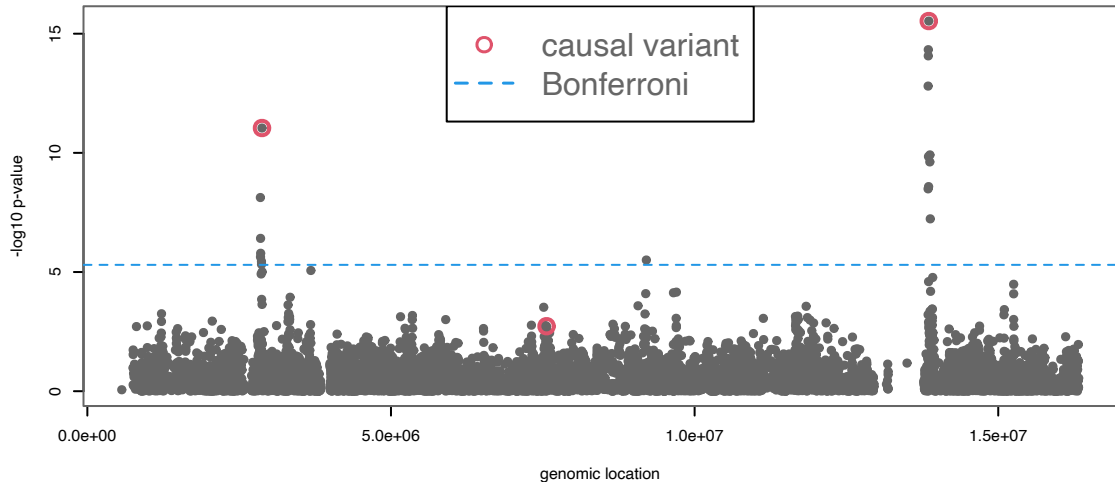
```
padj <- p.adjust(.gwas$pvalue)
sum(padj < .05)
```

```
## [1] 18
```

How many discovered vs. how many independent ones



How many discovered vs. how many independent ones



Adjusted p-value should control FWER only if...

```
padj <- p.adjust(gwas.p, method="bonferroni")
false.discoveries <- setdiff(which(padj < .05), causal.variants)
empirical.FDR <- length(false.discoveries) / sum(padj < .05)
print(empirical.FDR)
```

```
## [1] 0.8888889
```

```
padj <- p.adjust(gwas.p, method="BH")
false.discoveries <- setdiff(which(padj < .05), causal.variants)
empirical.FDR <- length(false.discoveries) / sum(padj < .05)
print(empirical.FDR)
```

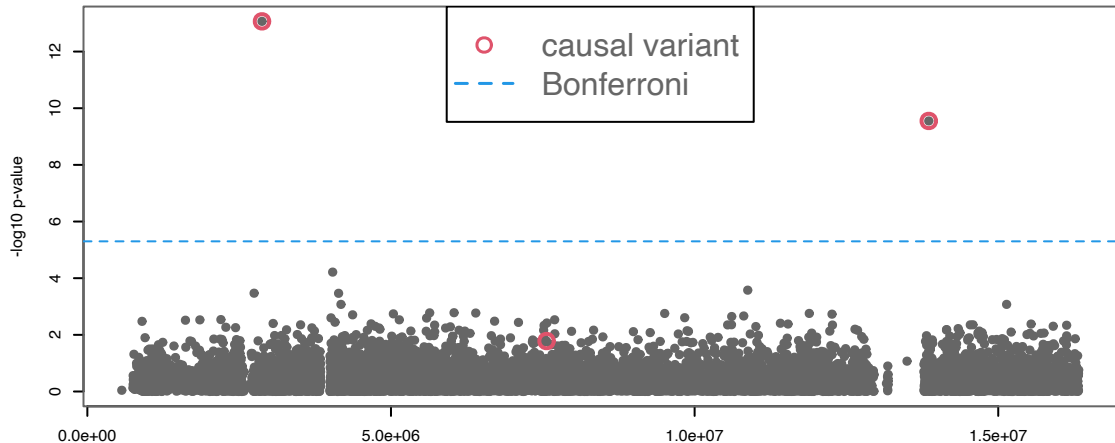
```
## [1] 0.9354839
```

What went wrong with GWAS?

What went wrong with GWAS?
Actually, everything is perfectly expected.

What if we do IID variant association studies?

- Instead of using a real-world genotype matrix, we could do the same exercise with IID 0,1,2 matrix (sample permutation).



Today's lecture

- 1 Mapping disease-specific locations in Genome
- 2 A toy example GWAS
- 3 How GWAS can be interpreted wrongfully**
- 4 Combining evidence from multiple studies

Exploratory Data Analysis of the 1KG data

In the 1KG data, we have 2490 rows/individuals and 1664852 columns/variants.

- We will use n to refer to the number of samples and p to the number of variants.
- Let's assume that we are mostly interested in biallelic (two alleles) variants/SNPs.

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- There are major (usually reference) and minor (usually effect) alleles. To save bits (in the early 2000s), geneticists used 0 for the major and 1 for the minor allele.
- So, each variant, we have a minor allele frequency (MAF), namely, f_j , for a variant j .

If each variant follows Binomial distribution

In Binomial distribution, with a MAF f_j , we have

- What will be a useful summary statistic for a variant j ?

Remark: Technically, the dosage (0,1,2) does not follow binomial distribution. Why? The underlying data generation process involves haplotypes (separating the maternal and paternal 0/1 counts) and dependency along the genomic axis.

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$$\hat{\mathbb{E}}[X_j] = 2f_j$$

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- What will be a useful summary statistic for a variant j ?

$$\hat{\mathbb{E}}[X_j] = 2f_j$$

- What is the variance of this variant?

$$\hat{\mathbb{V}}[X_j] = 2f_j(1 - f_j)$$

Remark: Technically, the dosage (0,1,2) does not follow binomial distribution. Why? The underlying data generation process involves haplotypes (separating the maternal and paternal 0/1 counts) and dependency along the genomic axis.

Variant-level MAF across individuals

We can easily calculate MAF using bigsnpr:

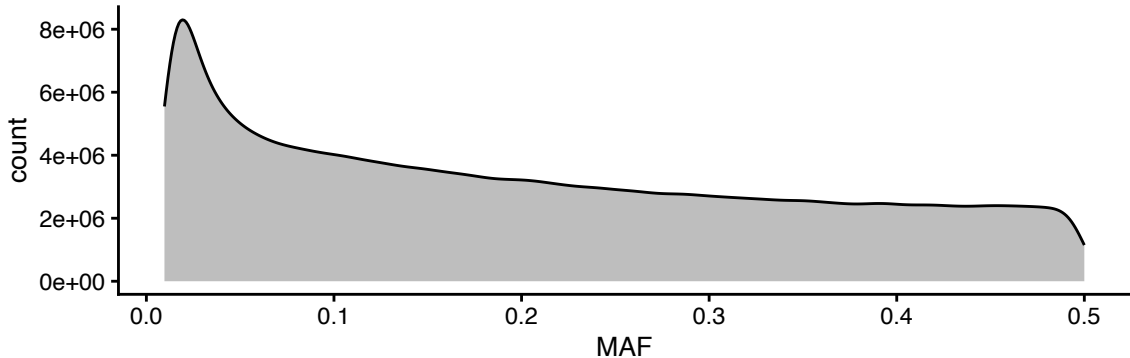
```
maf <- snp_MAF(data$genotypes, ncores=8)
```

- What are the x- and y-axis? Where is the peak?

Variant-level MAF across individuals

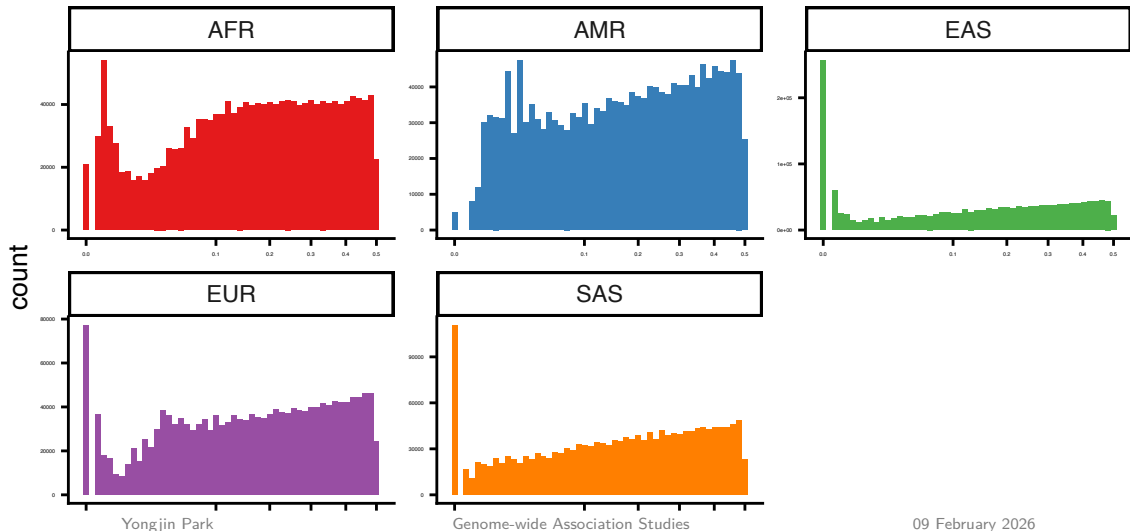
We can easily calculate MAF using bigsnpr:

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```



- What are the x- and y-axis? Where is the peak?

MAF distributions generally differ across ancestry groups (AG)

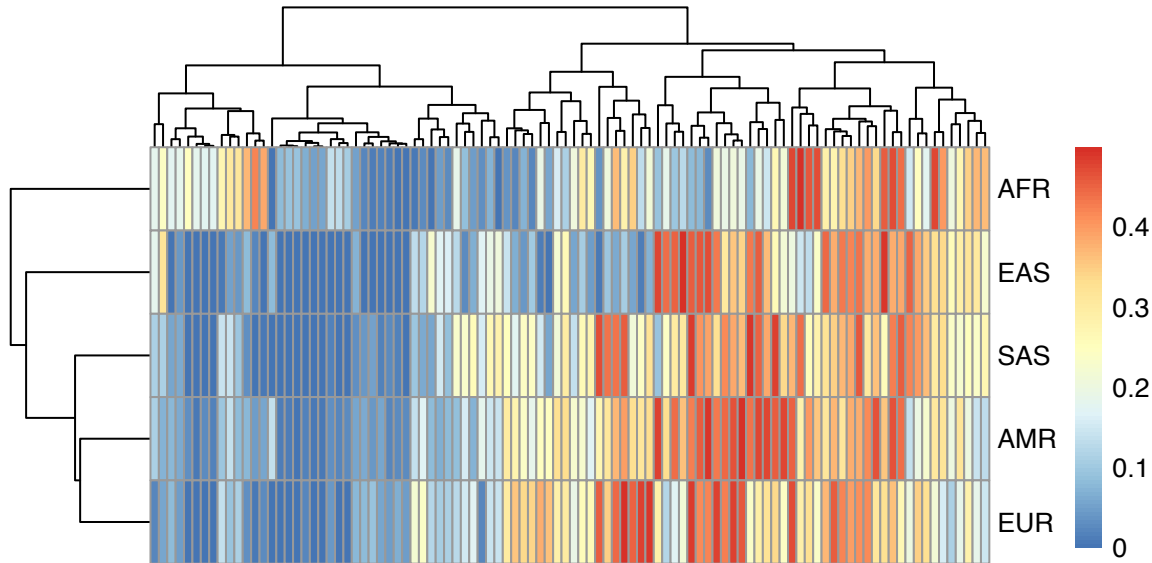


The same variant could have vastly different MAF values across AG

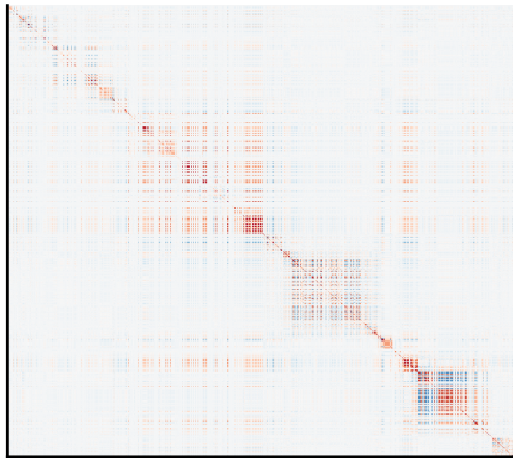
```
head(maf.mat)
```

##	AFR	AMR	EAS	EUR	SAS
##	<num>	<num>	<num>	<num>	<num>
## 1:	0.12576687	0.008645533	0.0000000	0.001988072	0.00000000
## 2:	0.28987730	0.263688761	0.2341270	0.161033797	0.22004132
## 3:	0.02607362	0.028818444	0.0000000	0.040755467	0.04028926
## 4:	0.35199387	0.243515850	0.2668651	0.128230616	0.19008264
## 5:	0.35199387	0.244956772	0.2371032	0.128230616	0.19008264
## 6:	0.45092025	0.175792507	0.1170635	0.130218688	0.16632231

Random 100 variants



The covariance matrices of X ? – between the variants



More discussions in the next lectures.

For a genotype matrix X ($n \times p$),

Linkage disequilibrium matrix

A $p \times p$ matrix:

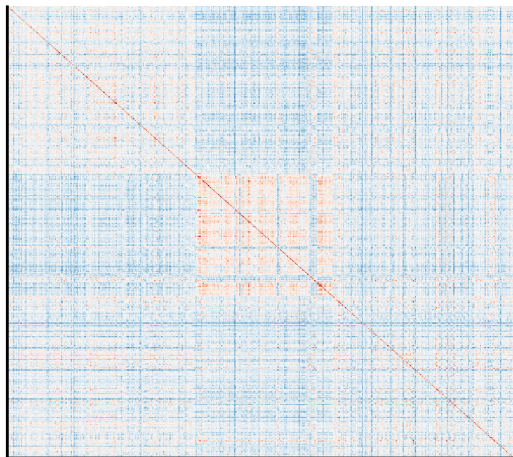
$$\hat{R} = \frac{1}{n} X^T X$$

given each column $\mathbf{x}_j$ standardized (mean 0, SD 1).

- n : individuals
- p : variants/SNPs

$$R_{ij} = \frac{1}{n} \sum_{r=1}^n X_{ri} X_{rj}$$

The covariance matrices of X ? – between the individuals



For a genotype matrix X ($n \times p$)

Kinship/genetic relatedness matrix

An $n \times n$ matrix:

$$\hat{K} = \frac{1}{p} X X^{\top}$$

given each row $\mathbf{x}_i$ standardized (mean 0, SD 1).

- n : individuals
- p : variants/SNPs

$$K_{ij} = \frac{1}{n} \sum_{a=1}^p X_{ia} X_{ja}$$

What would happen if there were AG-specific biases?

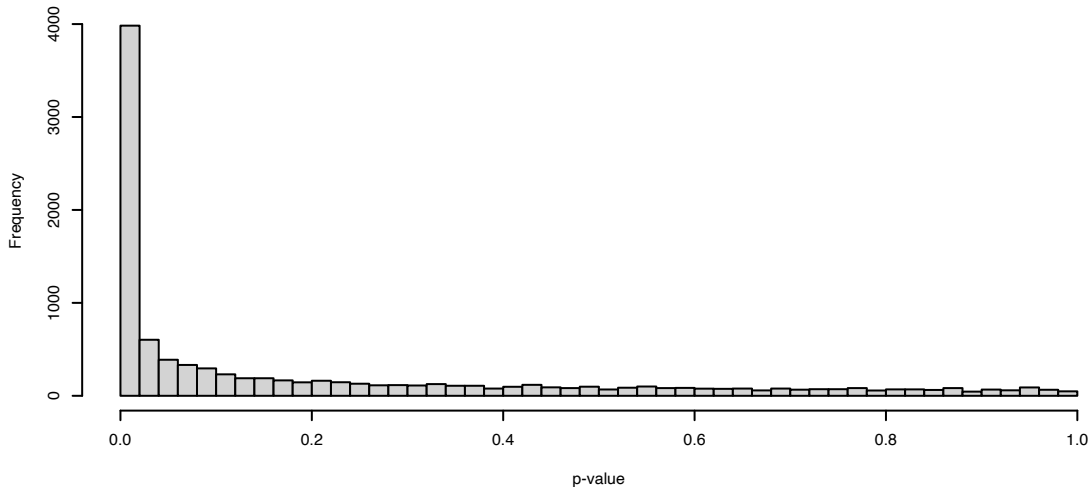
$$g \left(\begin{matrix} Y \\ \text{phenotype} \end{matrix} \right) \sim \underbrace{\sum_{k \in \text{causal variants}} X_k \beta_k}_{\text{true genetic effect}} + \underbrace{\sum_g \mathbf{U}_g \gamma_g}_g + \epsilon$$

ancestry group factors

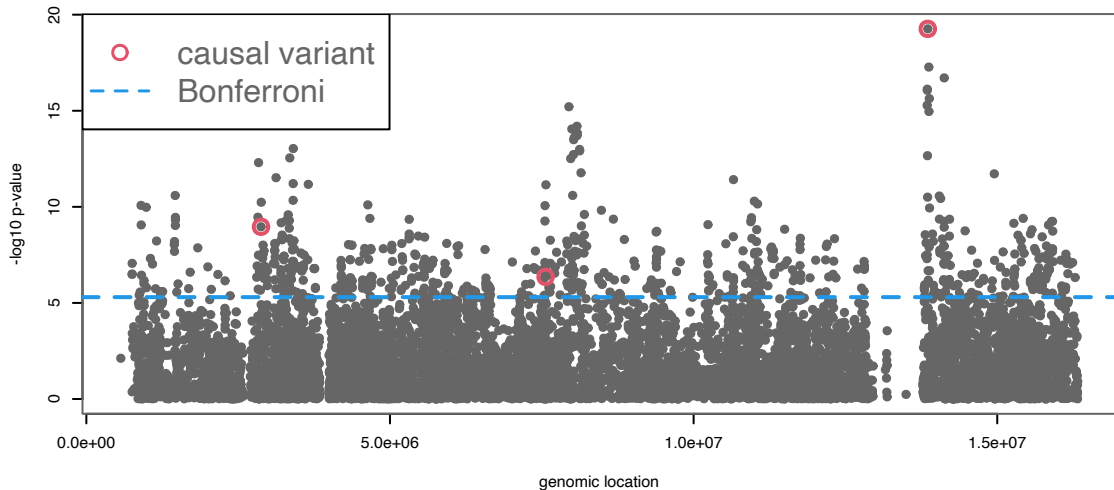
Do GWAS as usual by variant-by-variant t-test:

```
.gwas.biased <- col_t_welch(X[Y == 0, ], X[Y == 1, ])
```

Do we have that many significant ones?



Do we have that many significant ones?



How do we check group structures?

Common assumption:

- Only a small portion of variants are significantly associated. Why?
- A majority of p-values under the null will follow Uniform distribution.

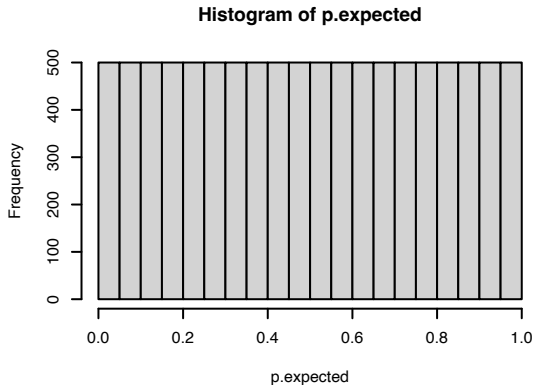
```
nn <- length(.gwas$pvalue)
p.expected <- seq(1, nn)/(nn+1)
```

How do we check group structures?

Common assumption:

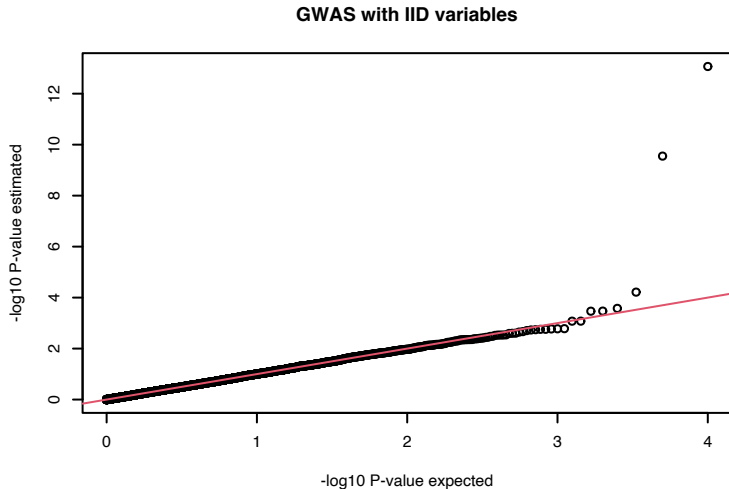
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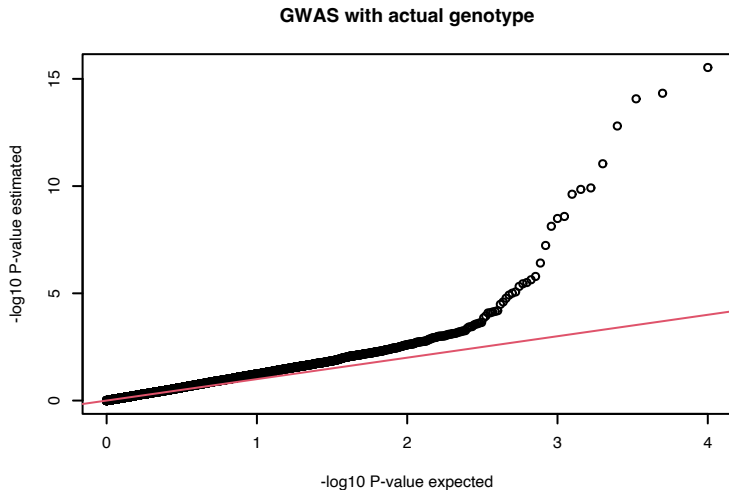
Quantile-quantile plot as a diagnostic tool (log-scale)

Use `qqplot(pvalue1, pvalue2)`



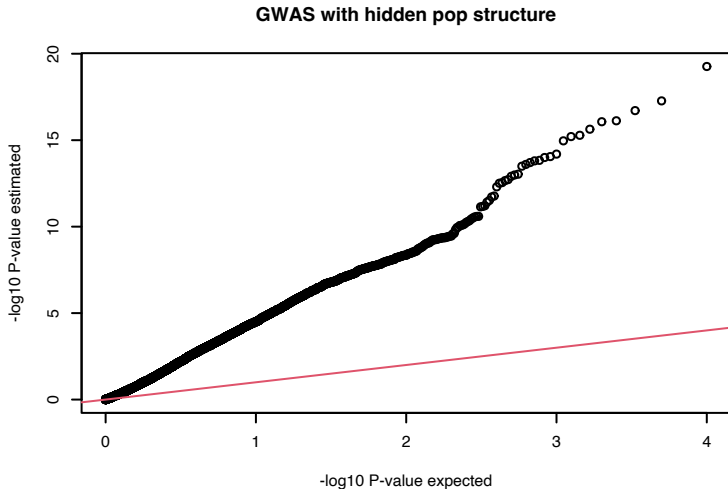
Quantile-quantile plot as a diagnostic tool (log-scale)

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Quantile-quantile plot as a diagnostic tool (log-scale)

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How can we ascertain ancestry group structures?

```
pca <- prcomp(t(X), rank=3) # rows
names(pca)

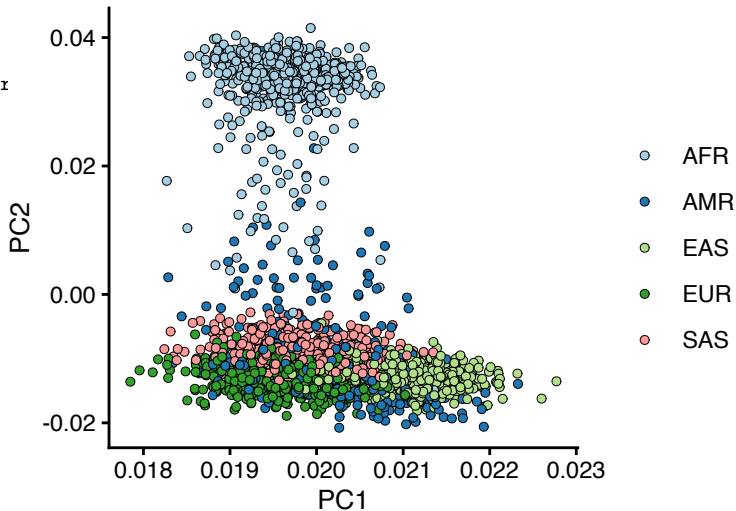
## [1] "sdev"      "rotation" "center"    "scale"     "x"

dim(pca$rotation)

## [1] 2490      3
```

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## [1] 2490      3
```



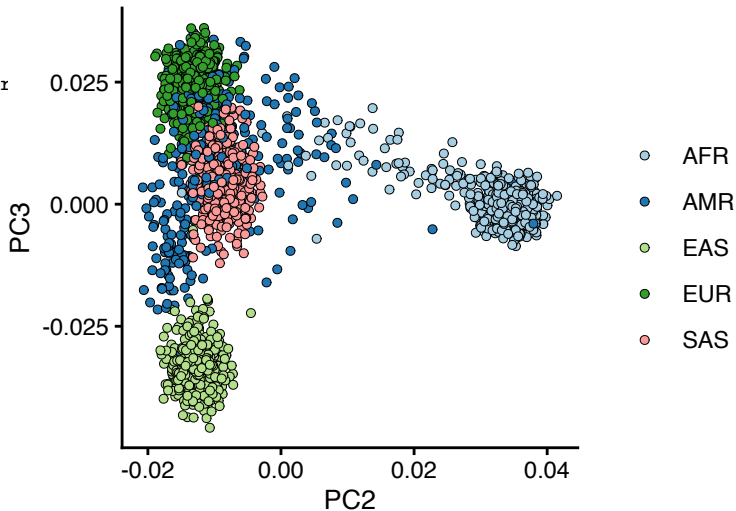
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names(pca)
```

```
## [1] "sdev"      "rotation" "center"
```

```
dim(pca$rotation)
```

```
## [1] 2490    3
```



Can we quickly fix this? A little bit...

```
.svd <- rsvd::rsvd(X, 10)
Y.adj <- lm(Y ~ .svd$u)$residual
.gwas.adj <- col_cor_pearson(X, Y.adj)
```

$$g \begin{pmatrix} Y \\ \text{phenotype} \end{pmatrix} \sim \underbrace{X_k \beta_k}_{\text{true genetic effect}} + \underbrace{\sum_g U_g \gamma_g}_{\text{estimate this by SVD}} + \epsilon$$

Recall: SVD captures principal components

$$X = UDV^{\top}$$

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What is this?

$$\frac{1}{n}X^{\top}X = \frac{1}{n}VDU^{\top}UDV^{\top} = \frac{1}{n}VD^2V^{\top}$$

variant x variant

Recall: SVD captures principal components

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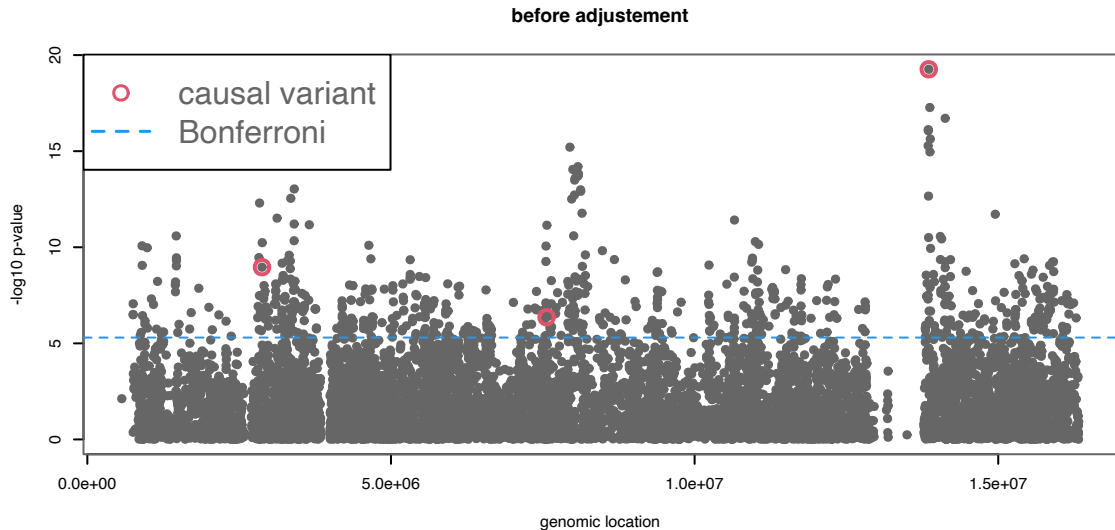
variant x variant

What is this?

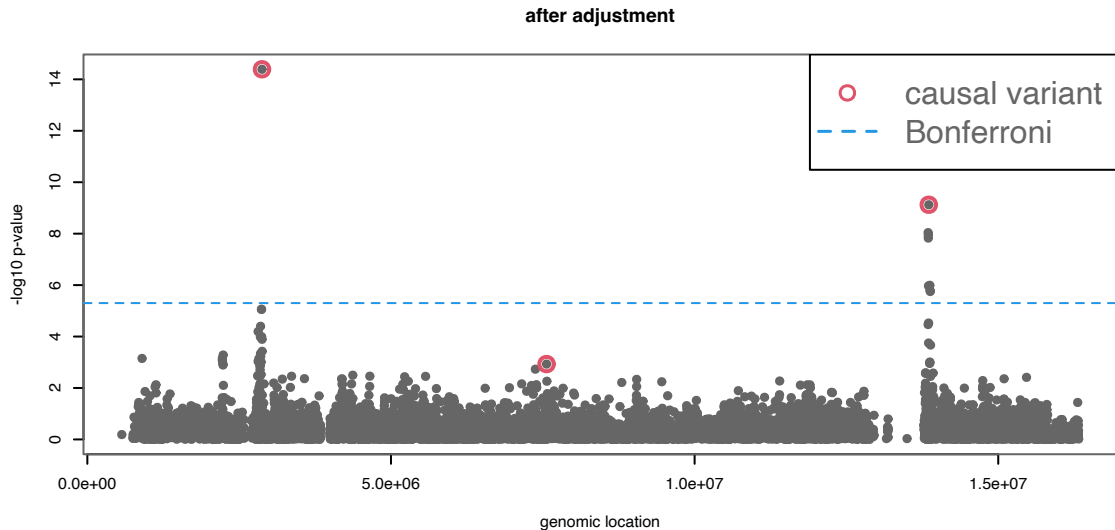
$$\frac{1}{n}XX^\top = \frac{1}{n}UDV^\top VDU^\top = \frac{1}{n}UD^2U^\top$$

sample x sample

Regressing out AG PCs can reduce biases

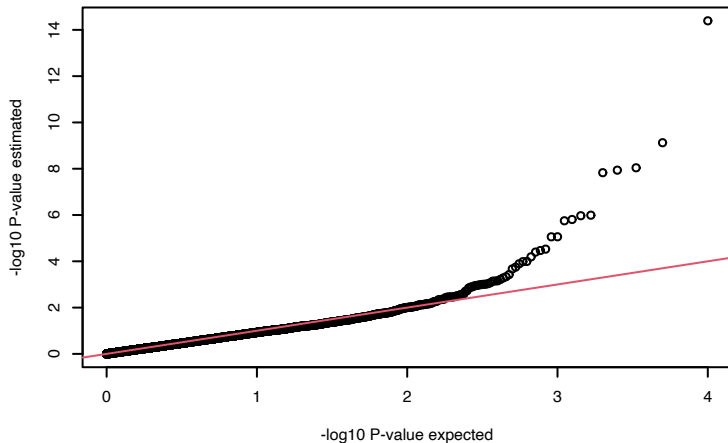


Regressing out AG PCs can reduce biases



Regressing out AG PCs can reduce biases

GWAS with actual genotype after adjustment

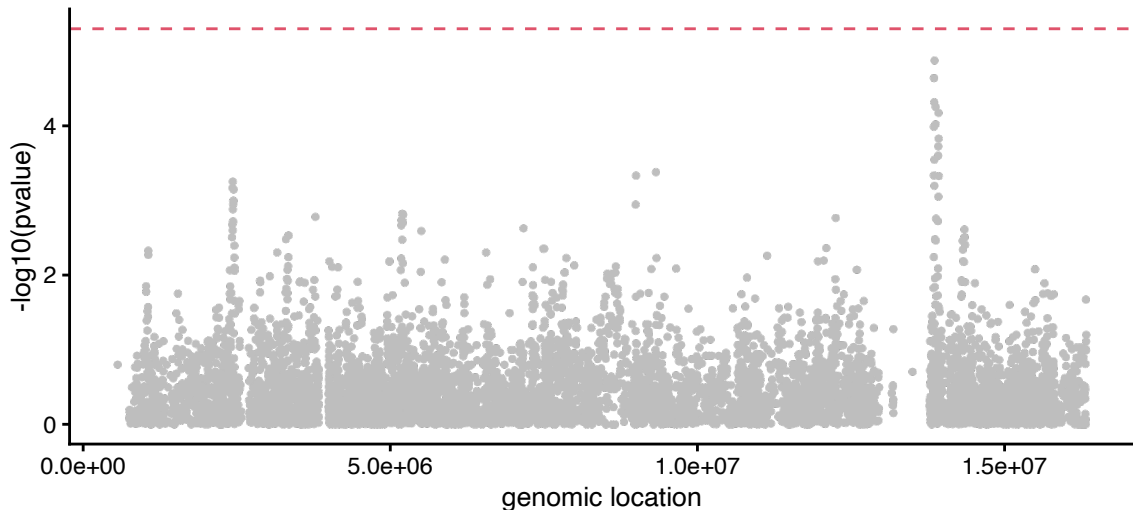


Today's lecture

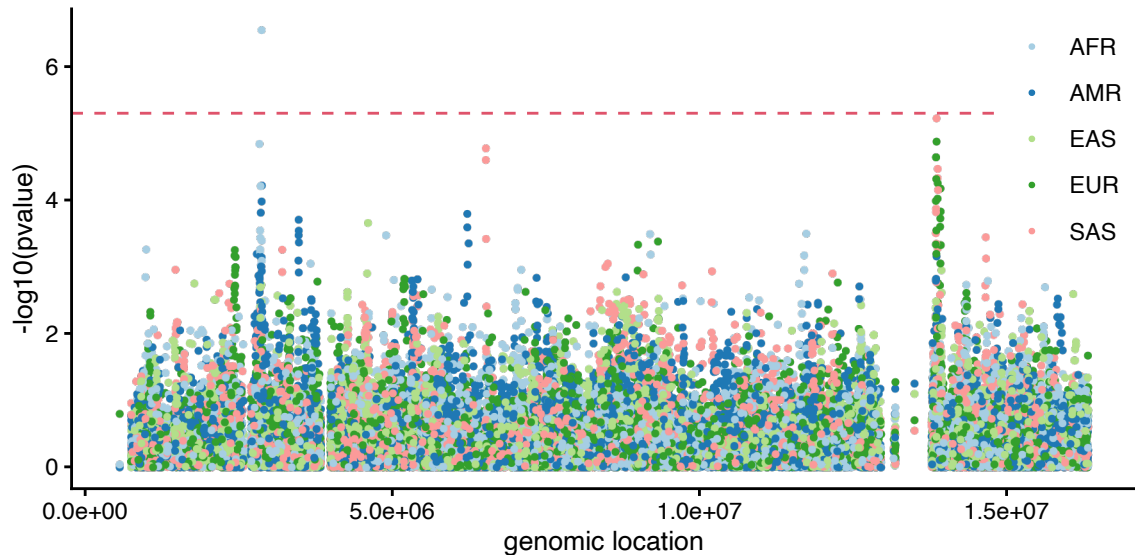
- 1 Mapping disease-specific locations in Genome
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Suppose we have GWAS statistics within each AG

GWAS in EUR (N=503)



Suppose we have GWAS statistics within each AG



Meta analysis: How can we combine multiple studies?

For each population/study k :

- $\hat{\beta}_{jk}$: effect size for a variant j on a study k
- $s_{jk} = \text{SE}(\beta_{jk})$: standard error for a variant j on a study k
- $w_{jk} = 1/s_{jk}^2$: inverse variance

Meta analysis by inverse variance weighting

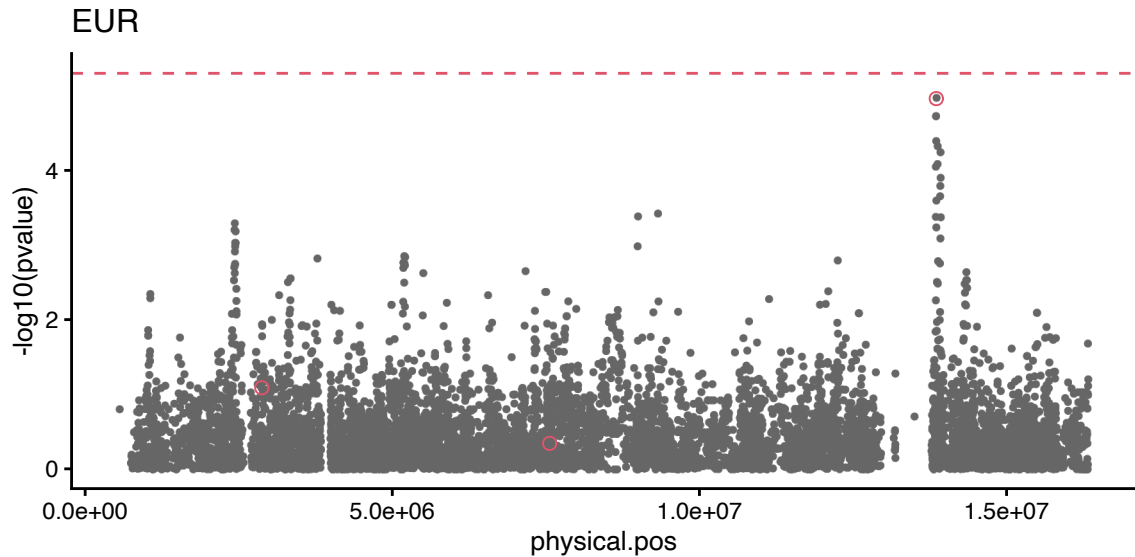
$$\hat{\beta}_j = \frac{\sum_k w_k \hat{\beta}_{jk}}{\sum_k w_k}, \quad \text{SE}(\hat{\beta}_j) = \frac{1}{\sqrt{\sum_k w_k}}$$

Meta analysis can boost statistical power

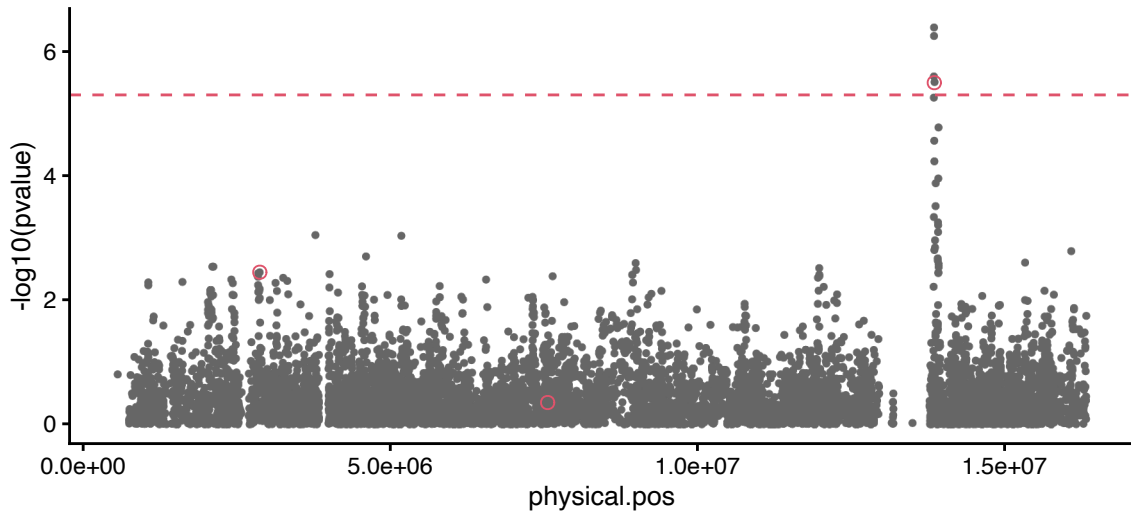
```
gwas.dt[, w := 1/(`stderr`^2 + 1e-16)]

meta.dt <-
  gwas.dt[!is.na(w),
    .(`mean` = sum(`mean.diff` * w)/sum(w),
      `se` = 1/sqrt(sum(w))),
    by = .(physical.pos)]

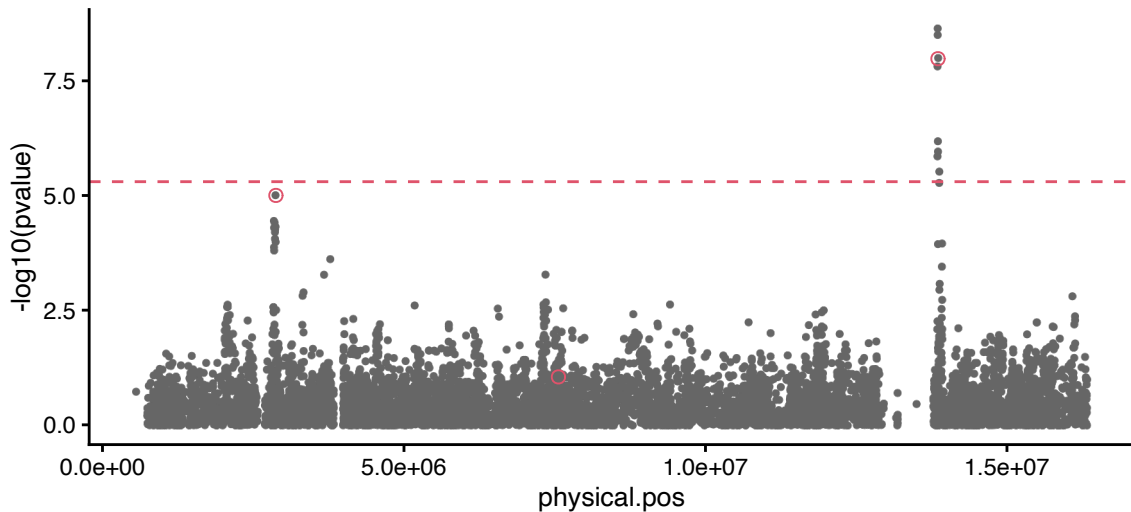
meta.dt[, `z` := `mean`/`se`]
meta.dt[, `pvalue` := 2*pnorm(abs(`z`), lower.tail=F)]
```



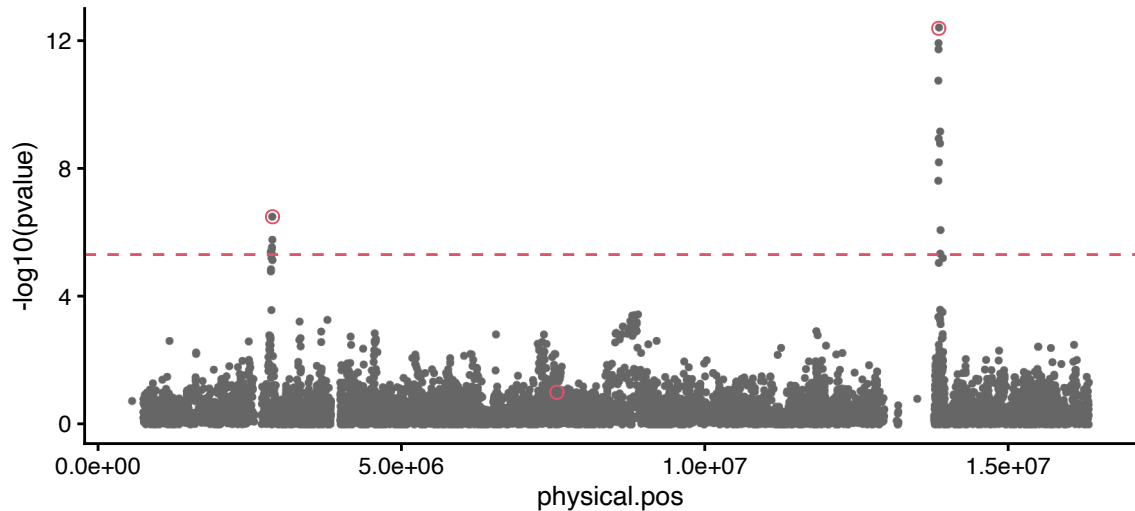
EUR + EAS



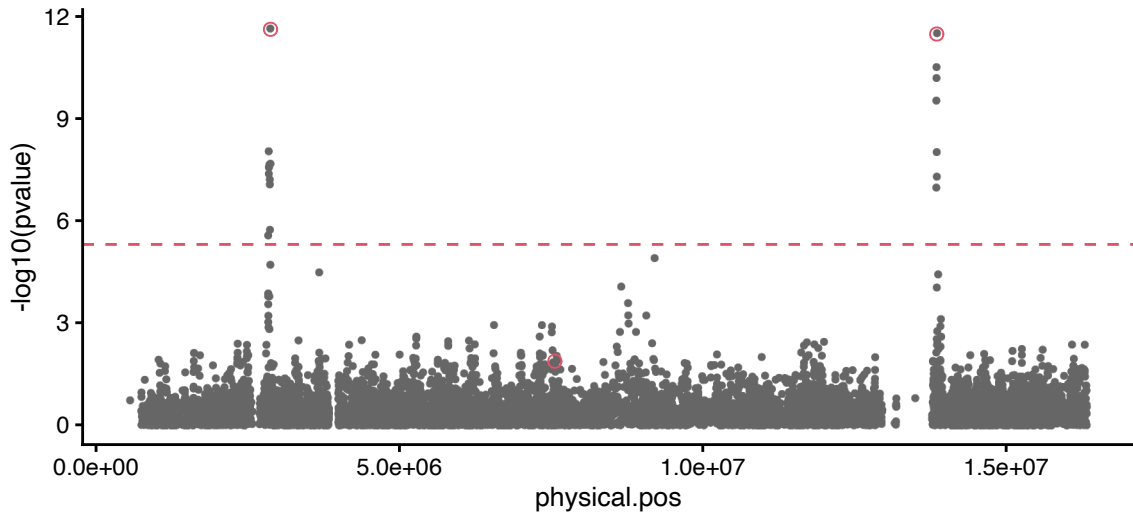
EUR + EAS + AMR



EUR + EAS + AMR + SAS



EUR + EAS + AMR + SAS + AFR



Common pitfalls of GWAS meta analysis

- Multiple GWAS results on similar yet slightly different phenotypes
- Different reference human genome
- Different array platforms (could cover different variants)
- Shared individuals
 - Some individuals can be shared among different studies
 - In UK Biobank data, many non-cancer individuals can be also non-neuro-degenerative disorder individuals.

Final pop quiz

What is the key assumption required for Bonferroni correction to be valid?

- Ⓐ All variants must have the same MAF
- Ⓑ Tests must be independent or positively correlated
- Ⓒ The sample size must be greater than 1000
- Ⓓ All variants must be in Hardy-Weinberg equilibrium

Final pop quiz

What is the primary purpose of including principal components (PCs) as covariates in GWAS?

- Ⓐ To increase the sample size
- Ⓑ To correct for population stratification
- Ⓒ To identify causal variants
- Ⓓ To improve genotyping accuracy

Final pop quiz

What is the purpose of a QQ (quantile-quantile) plot in GWAS?

- Ⓐ To identify which chromosome has the most associations
- Ⓑ To compare observed vs. expected p-value distributions under the null
- Ⓒ To visualize effect sizes
- Ⓓ To show LD between variants

Reference

- International HapMap Consortium. 2003. “The International HapMap Project.” *Nature* 426 (6968): 789–96.
- Manolio, Teri A, Francis S Collins, Nancy J Cox, et al. 2009. “Finding the Missing Heritability of Complex Diseases.” *Nature* 461 (7265): 747–53.
- Sally, Aylwyn, Julien Y Dutheil, Ladeana W Hillier, et al. 2012. “Insights into Hominid Evolution from the Gorilla Genome Sequence.” *Nature* 483 (7388): 169–75.